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United States Patent	12388323
Kind Code	B2
Date of Patent	August 12, 2025
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Control system for electric lawnmower and electric lawnmower including same

Abstract

An electric lawnmower control system can include a battery, an electric motor configured to rotatably drive a blade, a fan rotatably driven by the electric motor, a printed circuit board, a motor controller, and a housing. The printed circuit board can be fastened to the battery and include a heat sink that is located adjacent to the fan. The motor controller can be mounted on the circuit board, in electrical communication with the battery and the electric motor, thermally coupled to the heat sink, and configured selectively drive the electric motor with electric power supplied by the battery. The housing can include a first side, a second side, and a ventilation passage extending from the first side to the second side. The ventilation passage can be in fluid communication with the fan, and at least a portion of the heat sink is exposed to air flowing through the ventilation passage.

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Appl. No.:	17/571875
Filed:	January 10, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230223819 A1	Jul. 13, 2023

Publication Classification

Int. Cl.: H02K5/22 (20060101); A01D34/00 (20060101); A01D34/68 (20060101); A01D34/78 (20060101); A01D34/82 (20060101); H02K5/20 (20060101); H02K7/00 (20060101);

U.S. Cl.:

CPC **H02K9/14** (20130101); **A01D34/006** (20130101); **A01D34/6806** (20130101);
A01D34/78 (20130101); **A01D34/82** (20130101); **H02K5/20** (20130101); **H02K5/225**
(20130101); **H02K7/003** (20130101); **H02K7/14** (20130101); A01D2101/00 (20130101)

Field of Classification Search

CPC: H02K (5/225); H02K (5/20); H02K (9/14); A01D (34/001); A01D (34/006)

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Background/Summary

BACKGROUND

(1) The disclosed subject matter relates to an electric lawnmower. More particularly, the disclosed subject matter relates to methods and an apparatus that can manage power distribution among and cooling of electrical component(s) and/or electronic component(s) of the electric lawnmower.

(2) Lawnmowers can be configured as walk-behind lawnmowers or as riding lawnmowers. An electric lawnmower can be powered by one or more electric motors that can be supplied with power from an external power source via an electric wire or by an on-board power source such as one or more batteries. The electric lawnmower can include an electronic control unit (also referred to as a controller) and at least one operator input for the operator of the lawnmower to signal the controller how to operate the electric lawnmower. The electric lawnmower can include a fan that supplies cooling air that transfers heat from the controller, the electric motor(s), and the battery(ies) to the ambient environment.

SUMMARY

- (3) Some embodiments are directed to an electric lawnmower control system that can include a battery, an electric motor configured to rotatably drive a blade, a fan rotatably driven by the electric motor, a printed circuit board, a motor controller, and a housing. The printed circuit board can be fastened to the battery and include a heat sink that is located adjacent to the fan. The motor controller can be mounted on the circuit board, in electrical communication with the battery and the electric motor, thermally coupled to the heat sink, and configured selectively drive the electric motor with electric power supplied by the battery. The housing can include a first side, a second side, and a ventilation passage extending from the first side to the second side. The ventilation passage can be in fluid communication with the fan, and at least a portion of the heat sink is exposed to air flowing through the ventilation passage.
- (4) Some embodiments are directed to a control system for an electric lawnmower. The control assembly can include a battery, an electric motor configured to rotatably drive a blade, a fan rotatably driven by the electric motor, printed circuit board, a motor controller, and a housing. The battery assembly can include a battery case and at least one battery mounted in the battery case. The printed circuit board can be mounted on the battery and include a heat sink that is located adjacent to the fan. The motor controller can be mounted on the circuit board, in electrical communication with the battery and the electric motor, thermally coupled to the heat sink, and configured selectively drive the electric motor with electric power supplied by the battery. The housing can include a first side, a second side, and a ventilation passage extending from the first side to the second side. The ventilation passage is in fluid communication with the fan, and at least a portion of the heat sink is exposed to air flowing through the ventilation passage.
- (5) Some embodiments are directed to a control system for an electric lawnmower that can include a battery assembly, an electric motor configured to rotatably drive a blade, a fan rotatably driven by the electric motor, a motor controller, a heat sink, and a housing. The battery assembly can include a battery case and at least one battery mounted in the battery case. The motor controller can be in electrical communication with the battery and the electric motor and configured to selectively drive the electric motor with electric power supplied by the battery. The heat sink thermally coupled to the motor controller and located adjacent to the fan. The housing can include a first side, a second side, and a ventilation passage extending from the first side to the second side. The ventilation passage can be in fluid communication with the fan and at least a portion of the heat sink.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosed subject matter of the present application will now be described in more detail with reference to exemplary embodiments of the apparatus and method, given by way of example, and with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of a lawnmower made in accordance with principles of the disclosed subject matter.
- (3) FIG. 2 is a schematic side view of the lawnmower of FIG. 1.
- (4) FIG. 3 is a schematic top view of the lawnmower of FIG. 1.
- (5) FIG. 4 is an exploded schematic view of a control system of the lawnmower of FIG. 1.
- (6) FIG. 5 is schematic side view of the control system of the lawnmower of FIG. 1.
- (7) FIG. 6 is a schematic view of a first side of the printed circuit board of FIG. 5.
- (8) FIG. 7 is a schematic view of a second side of the printed circuit board of FIG. 5.
- (9) FIG. 8 is a schematic top view of a cooling assembly of the lawnmower of FIG. 1.
- (10) FIG. 9 is a schematic view of a comparative example of an electric lawnmower.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (11) A few inventive aspects of the disclosed embodiments are explained in detail below with

reference to the various figures. Exemplary embodiments are described to illustrate the disclosed subject matter, not to limit its scope, which is defined by the claims. Those of ordinary skill in the art will recognize a number of equivalent variations of the various features provided in the description that follows.

(12) An electric lawnmower can include a high voltage wire harness that electrically connects the battery(ies) to the controller, a wire harness that electrically connects the controller to the electric blade motor that drives the blade, and a wire harness that connects the controller to one or more operator inputs. Optionally, the lawnmower can include an additional wire harness that electrically connects the controller to an electric propulsion motor that propels the lawnmower along the ground. Each wire harness can include more than one electric wire and more than one electrical connector. The wire harnesses can add weight and space on the lawnmower. Further, each wire of each wire harness can be subject to damage. A damaged wire can prevent or hinder a desired operation of one or more components of the lawnmower. A damaged wire can cause a short circuit that can damage one or more components of the lawnmower.

(13) The high voltage wire harness can include one or more electric conductors that have relatively thick gauge. It can be difficult to solder the thick gauge wire(s) of the high voltage wire harness onto an electrical terminal connector. The high voltage wire harness can be difficult to manipulate during assembly and/or maintenance/repair due to the tight confines of the housing of the lawnmower. The electrical terminal connector can be relatively large in size, thereby adding to the difficulty in manipulating the high voltage wire harness during assembly and/or maintenance/repair.

(14) The lawnmower can vibrate each of the wire harnesses. Thus, it could be advantageous to provide each wire harness with a wire guide that can reduce at least some of the vibration of the wire harnesses. However, each wire guide can complicate the assembly of the lawnmower.

(15) Electric lawnmowers can be exposed to dust, debris and fluids, when in use, that can have an adverse impact on the controller. The cooling passage(s) can have a complex shape and/or at least one filter structure that reduces the amount of dust, debris and fluids in the air drawn into the lawnmower by the fan. The fan can be operated by the electric motor that drives the cutting blade. The complex shape of the cooling passage(s) and the filters can cause frictional losses and pressure drops that, in turn, can cause the electric motor to consume a relatively high amount of the battery's power. This relatively high power consumption can reduce the running time of the lawnmower for a given charge of the battery.

(16) Accordingly, it can be advantageous to provide an electric lawnmower that can omit at least one high voltage wire harness while maintaining the operations and power distribution of a lawnmower. It can also be advantageous to provide an electric lawnmower with a cooling system that provides sufficient air flowing through the lawnmower while minimizing the power drawn by the fan to move the air through the lawnmower.

(17) FIG. 1 illustrates an embodiment of a lawnmower **10** made in accordance with principles of the disclosed subject matter. The lawnmower **10** can extend in an X-direction, a Y-direction, and a Z-direction, which directions are orthogonal to each other. The lawnmower **10** can include a cutter housing **12**, a pair of front wheels **14**, a pair of rear wheels **16**, a handle **18** and a control system **20**. The rear wheel **16** on the right side of the lawnmower is obstructed from view in FIG. 1 by the cutter housing **12**. FIG. 3 schematically illustrates the right rear wheel **16**.

(18) Referring to FIG. 2, the lawnmower **10** can include a blade **22** and a blade shaft **24** connected to each of the blade **22** and the control system **20**. The control system **20** can be configured to selectively rotate the blade shaft **24** and the blade **22** in the cutter housing **12** about a blade axis A. The blade shaft **24** can be referred to as a component of the control system **20**. Alternatively, the blade shaft **24** can be referred to as a component that is connected to and driven by the control system **20**.

(19) Returning to FIG. 1, the cutter housing **12** can be referred to as a mower deck or as a cutter

deck or as a deck. The cutter housing **12** can include an opening at a rear end **26** of the cutter housing **12**. The lawnmower **10** can include a collection bag **36** that can be selectively attached to and detached from the rear end **26**. The opening is omitted for simplicity and clarity of the drawing. The collection bag **36** can be in communication with the opening such that vegetation clippings produced by the blade **22** can be collected in the collection bag **36**.

(20) FIGS. **2-4** schematically illustrate exemplary components of the control system **20**. The control system **20** can include a housing **28**, a battery pack **30**, a blade motor **32**, a blade motor controller **34**, an operator input assembly **38** and a cooling assembly **40**. The cooling assembly **40** is omitted from FIGS. **3** and **4** for simplicity and clarity of the drawing figures.

(21) Referring to FIGS. **1-3** collectively, the housing **28** can contain the battery pack **30**, the blade motor **32**, the blade motor controller **34** and the cooling assembly **40**. The housing **28** can include a front end **56**, a rear end **58**, left side **60**, a right side **62** and a top surface **64**. The sides **60**, **62** can extend along all of the battery pack **30**, the blade motor **32**, the blade motor controller **34** and a printed circuit board **54** on which the blade motor controller **34** is mounted.

(22) The battery pack **30** can be located in the housing **28** at a position that is adjacent to a front end **54** of the housing **28**. Referring to FIG. **2**, the battery pack **30** can include at least one battery cell **42** and a case **44** that houses the at least one battery cell **42**. The battery cell(s) **42** can be configured to store electricity and supply electricity to the blade motor **32**.

(23) The blade motor **32** can be a direct current electric motor or an alternating current electric motor. Embodiments can include a blade motor **32** that is configured as a direct current outer rotor motor that includes an inner stator and an outer rotor. The blade motor **32** can include one or more sensors that provide the blade motor controller **34** with information regarding the temperature, rotational speed, power output, etc., of the blade motor **32**. The outer rotor of the blade motor **32** can be directly connected to the blade shaft **24** in any appropriate manner such that the blade motor **32** can cause the blade shaft **24** to rotate.

(24) The blade motor controller **34** can also be referred to as an electronic control unit (ECU) or as a central processing unit. The blade controller **34** can include a processor and a memory storage device. The blade motor controller **34** can be configured with hardware, with or without software, to perform the assigned task(s).

(25) The blade motor controller **34** can be in electrical communication with each of the battery pack **30**, the blade motor **32** and the operator input assembly **38**. The blade motor controller **34** can be configured to convert power from the battery pack **30** into output power supplied to the blade motor **32**. The blade motor controller **34** can be configured to monitor the operational conditions of the blade motor **32** and the battery pack **30**. The blade motor controller **34** can be configured to control the voltage and/or current output by the battery pack **30** based on the operational conditions of the blade motor **32** and the battery pack **30** switching one or more power transistors to adjust the supply of electrical power to the blade motor **32**.

(26) The blade motor controller **34** can also be configured to control the voltage or current output by the battery pack **30**, and to supply the voltage or current to the blade motor **32** using the one or more power transistors based on one or more inputs to the operator input assembly **38** by the operator of the electric lawnmower **10**.

(27) The blade motor controller **34** can be configured to initiate, adjust or terminate supply of voltage or current from the battery pack **30** to the blade motor **32** based on inputs received from the operator input assembly **38**, the battery pack **30** and the blade motor **32**. The blade motor controller **34** can also be configured to regulate the charging of the battery cell(s) **42** of the battery pack **30**.

(28) Referring to FIGS. **2-4** collectively, the control system **20** can include a battery connector **46**, a first controller connector **48**, a second controller connector **50**, a motor connector **52** and the printed circuit board **54**. Each of the connectors **46**, **48**, **50**, **52** can be an electrical connector that includes a housing and at least one electrically conductive terminal contained in the housing. The battery connector **46** can be mounted on or integrally formed with the case **44**. Referring to FIG. **4**,

the blade motor controller **34**, the first controller connector **48** and the second controller connector **50** can be directly mounted on the printed circuit board **54**. As a result, the printed circuit board **54** can be directly connected to the battery pack **30** and a large gauge wire conductor for conducting the high voltage from the battery pack **30** to the blade motor controller **32** can be omitted. This can simplify assembly, reduce weight of the lawnmower **10**, reduce the manufacturing cost and part cost of the lawnmower, and further fortify the lawnmower **10** against hindered operation or damage due to a short circuit. As will be described in further detail below, directly connecting the printed circuit board **54** to the battery pack **30** can improve the cooling efficiency of the blade motor controller **34** and can reduce the amount of battery power that is consumed by the cooling assembly **40**.

(29) Referring to FIG. **4**, the control system **20** can include a plurality of input connectors **66**, **68**, **70**, **72**, and a plurality of leads **74**. Each of the input connectors **66**, **68**, **70**, **72** can include a connector housing, at least one terminal contained within the connector housing, and at least one wire **76** electrically connected to the at least one terminal and extending out of the housing. The lead **74** can be part of the printed circuit of the printed circuit board **54**. The wires **76** of the connectors **66**, **68**, **70**, **72** can be soldered onto a respective one of the leads **74**. The leads **74** and the connectors **66**, **68**, **70**, **72** can electrically connect the blade motor controller **34** to the operator input assembly **38**.

(30) The control system **20** can include a plurality of wire harnesses **78**, **80**, **82** that can electrically connect the blade motor controller **34** to the operator input assembly **38**.

(31) The board wire harness **78** can include a first board harness connector **84**, a second board harness connector **86** and a plurality of first wires **88**. The first wires **88** can be electrically connected to each of the connectors **84**, **86**. The first board connector **84** can be physically and electrically connected to the first input connector **66**.

(32) The blade wire harness **80** can include a first blade harness connector **90**, a second blade harness connector **92**, a third blade harness connector **94** and a plurality of second wires **96**. The second wires **96** can be electrically connected to the connectors **90**, **92**, **94** as shown in FIG. **4**. The first blade harness connector **90** can be physically and electrically connected to the second input connector **68** and the second blade harness connector **92** can be physically and electrically connected to the third input connector **70**.

(33) The fold wire harness **82** can include a first fold harness connector **98**, a second fold harness connector **100**, a third fold harness connector **102** and a plurality of third wires **104**. The third wires **104** can be electrically connected to each of the connectors **98**, **100**, **102** as shown in FIG. **4**. The first fold harness connector **98** can be physically and electrically connected to the fourth input connector **72**.

(34) Referring to FIG. **1**, the operator input assembly **38** can include a blade operation input **106** and a handle operation input **108**. The inputs **106**, **108** can be any appropriate structure that can permit the operator of the lawnmower **10** to signal to blade motor controller **34** the operator's intention regarding the operation of the blade motor **32**. In FIG. **1**, the inputs **106**, **108** are shown as levers that can be pivotally connected to the handle **18**. Instead of the pivoting levers shown in FIG. **1**, the inputs **106**, **108** can be configured as a push button, a dial, a capacitive sensor or any other appropriate input device.

(35) Returning to FIG. **4**, the operator input assembly **38** can include an indicator board **110**, a blade switch **112**, a handle switch **114** and a folding switch **116**. The indicator board **110** can include at least one visual message device **118** that can provide an operator the lawnmower **10** with information regarding the operational status of the lawnmower **10**. The visual message device **118** can be a liquid crystal display and/or one or more status lights that are controlled based on one or more signals the visual message device **118** receives from the blade motor controller **34**.

(36) The blade switch **112** can be configured to transmit to the blade motor controller **34** a blade ON signal when the operator actuates the blade input **106** and a blade OFF signal when the

operator releases the blade input **106**. Based on a signal received from the blade switch **112**, the blade motor controller **34** can be configured to selectively turn off and on electrical power to the blade motor **32**.

(37) The handle switch **114** can be configured to transmit a ready signal when the operator actuates the handle input **108** and a sleep signal when the operator releases both of the blade input **106** the handle input **108**. Based on the signal received from the handle switch **114**, the blade motor controller **34** can be configured to wake up from or enter a sleep state.

(38) The folding switch **116** can be configured to selectively turn on and off electrical communication between the switches **112**, **114** and the blade motor controller **34**. The folding switch **116** can be configured to terminate electrical communication between the switches **112**, **114** and the blade motor controller **34** when the handle **18** is folded toward the housing **28** and permit electrical communication between the switches **112**, **114** and the blade motor controller **34** when the handle **18** is unfolded away from the housing **28** and locked in the use position that is shown in FIG. 1.

(39) The operator input assembly **38** can include an indicator connector **120**, a handle connector **122**, a first fold connector **124** and a second fold connector **126**.

(40) The indicator connector **120** can be electrically connected to the indicator board **110** and the visual message device **118**. The indicator connector **120** can be physically and electrically connected the second board harness connector **86**.

(41) The handle connector **122** can be electrically connected to each of the blade switch **112** and the handle switch **114**. The handle connector **122** can be physically and electrically connected to the third blade harness connector **94**.

(42) The fold connectors **124**, **126** can be electrically connected to the folding switch **116**. The first and second fold connectors **124**, **126** can be physically and electrically connected to the second and third fold harness connectors **100**, **102**, respectively.

(43) FIG. 3 depicts the input connectors **66**, **68**, **70**, **72** collectively as an input connector **128**, and the wire harnesses **78**, **80**, **82** collectively as a lawnmower wire harness **130** that includes a harness connector **132** that is electrically connected to the input connector **128**.

(44) FIGS. 5-7 show further details of the controller connectors **48**, **50**. The printed circuit board **54** can include a first side **134** and a second side **136**. The first controller connector **48** can be mounted on the first side **134** and the second controller connector **50** can be mounted on the second side **136**.

(45) Referring to FIGS. 5 and 7, the first controller connector **48** can include a terminal housing **138**, mounting flange **140** and a plurality of electrical terminals **142** mounted in the terminal housing **138**. The flange **140** can be a continuous flange that extends across and beyond the terminal housing **138**, or the flange **140** can include two separate flanges, one on each side of the terminal housing **138**. The flange(s) **140** can be connected to or integrally formed with the terminal housing **138**. A pair of screws **144** can connect the first controller terminal **48** to the first side **134** of the printed circuit board **54**. The terminals **142** can exit a rear end **146** of the terminal housing **138** and extend toward the printed circuit board **54**. The terminals **142** can be electrically connected by solder **148** to a respective lead or via of the printed circuit board **54**.

(46) Referring to FIGS. 4 and 6, the second controller connector **50** can include two separate connector housings **150**, **152** and a plurality of electrical terminals **154** mounted in the connector housings **150**, **152**. The printed circuit board **54** can include a plurality of leads **155**. The terminals **154** can be connected to the leads **155** of the printed circuit board **54** by solder.

(47) Returning to FIGS. 2, 3, 5 and 6, the cooling assembly **40** of the control system **20** can include a heat sink **156** and a fan **158**. The heat sink **156** can be thermally coupled to the blade motor controller **34** such that heat generated by the blade motor controller **34** is transferred by thermal conduction to the heat sink **156**. Mounting the printed circuit board **54** directly to the battery pack **30** can permit the printed circuit board **54** to be mounted in a space that separates the fan **158** and the battery pack **30**. This feature can permit the heat sink **156** to be positioned in close proximity to

the fan **158**. The close proximity of the heat sink **156** to the fan **158** can reduce the length of a cooling duct that directs air over the heat sink **156**. Accordingly, the power consumption by the fan **158** can be reduced and the charge of the battery pack **30** can be extended as compared to a layout such as that shown in the comparative example of FIG. **9**.

(48) Referring to FIG. **8**, the cooling system **40** can include a fan housing **160**, an inlet **162**, an outlet **164**, a first ventilation passage **166** and a second ventilation passage **168**.

(49) The fan housing **160** can be a portion of the housing **28**. In an alternative embodiment, the fan housing **160** can be a separate component from the housing **28**. The fan **156** can be located inside the fan housing **160**.

(50) Referring to FIGS. **1** and **8**, the inlet **162** can be an opening formed on the right side **62** of the housing **28** and the outlet **164** can be an opening formed on the left side **60** of the housing **28**. As shown in FIG. **1**, the outlet **164** can include a screen or mesh. Although not viewable in FIG. **1**, the inlet **162** can also include a screen or mesh as shown for the outlet **164**.

(51) The fan housing **160** can be centered between the left side **60** and the right side **62** of the housing **28** in the X-direction of the lawnmower **10**. Placing the inlet **162** and the outlet **164** on the right side **62** and the left side **64**, respectively, can allow the length of each of the passages **166**, **168** to be minimized with respect to other locations of the inlet **162** and **164**, such as at the front end **56** and/or the rear end **58** of the housing **28**.

(52) The inlet **162** can be located between the fan **158** and the battery pack **30** in the Y-direction of the lawnmower **10**. The first ventilation passage **166** can extend from the inlet **162** in the X-direction of the lawnmower **10** and into the space between the fan **158** and the battery pack **30**. Connecting the printed circuit board **54** directly to the battery pack **30** can position the heat sink **156** in the space between the fan **158** and the battery pack **30**. The heat sink **156** can intersect the first ventilation passage **166** such that at least a portion of the heat sink **156** extends into the first ventilation passages **166**, as shown in FIGS. **2** and **8**. The relatively short length and substantially straight path of the first ventilation passage **166** can reduce the frictional losses of the air flowing through the first ventilation passage **166**. Accordingly, the heat sink **156** can be efficiently cooled with a minimum consumption by the fan **158** of power from the battery pack **30**.

(53) The first ventilation passage **166** can extend from the inlet **162** to the fan housing **160** and can be in fluid communication with each of the inlet **162** and the fan housing **160** such that, during operation of the fan **158**, air can flow from outside of the housing **28** and enter the fan housing **160** via the inlet **162** and the first ventilation passage **166**.

(54) The second ventilation passage **168** can extend from the fan housing **160** to the outlet **164** and can be in fluid communication with each of the fan housing **160** and the outlet **164** such that, during operation of the fan **158**, air can flow from the fan housing **160** and exit the outlet **164** via the second ventilation passage **168**. The second ventilation passage **168** can form a curved path for the air flowing through the passage **168** and exiting the housing **28** via the outlet **164**.

(55) FIG. **9** shows a comparative example lawnmower **500** that includes a battery pack **502**, a blade motor **504**, a fan **506** and printed circuit board **508**. The printed circuit board **508** can include a heat sink **510** and a controller (not shown) that operates the blade motor **504**. The printed circuit board **508** and the heat sink **510** can be located between the fan **506** and the rear wheel **512**. That is, the printed circuit board **508** and the heat sink **510** are spaced away from a space that extends between the battery pack **502** and the fan **506**. A heavy gauge wire harness (not shown) can extend from the printed circuit board **508** to the battery pack **502**. In addition to having the assembly, weight, cost and wire harness damage issues discussed above with respect to the heavy gauge wire harness, the printed circuit board **508** can be at an increased elevation relative to the cutter housing **514**. The housing **28** of the lawn mower **10** is shown in phantom in FIG. **9** and superimposed on the lawnmower **500**. In addition to improving the cooling efficiency and omitting a large gauge wire conductor, locating the printed circuit board **56** and the heat sink **156** of FIGS. **2-8** in the space between the battery pack **30** and the fan **158** can allow the height of the housing **28** to be reduced

by a distance H as indicated in FIG. 9. Thus, the lawnmower **10** of FIGS. **1-8** can provide a lower profile—and reduced storage size when the handle **18** is folded onto the housing **28**.

(56) While certain embodiments of the invention are described above, it should be understood that the invention can be embodied and configured in many different ways without departing from the spirit and scope of the invention.

(57) For example, embodiments are disclosed above in the context of walk-behind lawnmower shown in FIG. **1**. However, exemplary embodiments can include or any type lawnmower described above. For example, the lawnmower can be configured as a ride-on lawnmower. In another alternate embodiment, the lawnmower can be a self-propelled lawnmower to which a stand-on sulky or a sit-on sulky can be attached. However, the above alternative embodiments are merely provided for exemplary purposes, and as indicated above, embodiments are intended to cover any type of type of lawnmower disclosed above.

(58) Alternate embodiments can include a reversed location of the inlet **162** and the outlet **164** such that the inlet **162** is on the left side **60** and the outlet is on the right side **62** of the housing **28**.

(59) Alternate embodiments can include the outlet **164** located between the battery pack **30** and the fan **158** in the Y-direction of the lawnmower **10**, the second ventilation passage can be substantially straight in the X-direction of the lawnmower **10**, and at least a portion of the heat sink **156** can extend into the second ventilation passage **168**.

(60) Alternate embodiments can include the fan housing **160** and the fan **158** located below the blade motor **32**.

(61) Instead of a wire extending from the blade motor **32** to the motor connector **52** as described above, the motor connector **52** can be mounted on or integrally formed with the housing of the blade motor **32**.

(62) Instead of the handle operation input **108**, the operator input assembly **38** can include a push button that can actuate the handle switch **114** to cause the handle switch to transmit the output signal in the manner described above.

(63) The operator input assembly **38** can include a main controller mounted on the handle. Instead of the blade motor controller **32**, the main controller can be configured to receive the signals from the switches **112**, **114**, **116**. Based on a signal received from the handle switch **114** and/or the folding switch **116**, the main controller can be configured to wake up from a sleep state, wake up the blade motor controller **34** from a sleep state, and/or selectively open and close a switch that selectively turns off and on, respectively, electrical output by the battery pack **30**. Based on a signal received from the blade switch **112**, the main controller **34** can be configured to signal the blade motor controller **34** to selectively turns off and on electrical power to the blade motor **32**.

Claims

1. An electric lawnmower control system comprising: a battery; an electric motor configured to rotatably drive a blade; a fan housing; a fan located in the fan housing and rotatably driven by the electric motor; a printed circuit board fastened to the battery and including a heat sink that is located adjacent to the fan; a motor controller mounted on the circuit board, in electrical communication with the battery and the electric motor, thermally coupled to the heat sink, and configured to selectively drive the electric motor with electric power supplied by the battery; and a housing including a first side, a second side, and a ventilation passage extending from the first side to the second side, the ventilation passage is in fluid communication with the fan and the fan housing, and at least a portion of the heat sink is exposed to air flowing through the ventilation passage, wherein the heat sink extends into the fan housing.
2. The electric lawnmower control system according to claim 1, further comprising: a battery connector mounted on the battery and in electrical communication with the battery; and a controller connector mounted on the printed circuit board, physically and electrically connected to the battery

connector, and in electrical communication with the motor controller.

3. The electric lawnmower control system according to claim 2, wherein the controller connector includes, a terminal housing, a plurality of terminals mounted in the terminal housing, soldered to the printed circuit board, and in electrical communication with the motor controller, and a mounting flange that extends from the terminal housing and is fastened to the printed circuit board by at least one screw.

4. The electric lawnmower control system according to claim 1, further comprising: a motor connector electrically connected to the electric motor; and a controller connector mounted on the printed circuit board, physically and electrically connected to the motor connector, and in electrical communication with the motor controller.

5. The electric lawnmower control system according to claim 4, wherein the controller connector includes, a terminal housing, a plurality of terminals mounted in the terminal housing, soldered onto the printed circuit board, and in electrical communication with the motor controller, and a mounting flange that extends from the terminal housing and is fastened to the printed circuit board by at least one screw.

6. The electric lawnmower control system according to claim 1, further comprising: a first connector in electrical communication with the motor controller; a second connector in electrical communication with the motor controller; and a motor wire harness including at least one motor connector and a plurality of wires electrically connected to the motor and the motor connector, the motor connector is physically and electrically connected to the second connector, wherein the printed circuit board includes a first side that faces the battery and a second side that faces the fan and the electric motor, the second connector is mounted on the second side of the printed circuit board, the first connector is mounted on the first side of the printed circuit board, and the battery includes a battery connector that is physically and electrically connected to the first connector.

7. The electric lawnmower control system according to claim 1, wherein the fan and the electric motor are separated from the battery by a space, and the printed circuit board is located in the space.

8. The electric lawnmower control system according to claim 1, wherein the printed circuit board is located between the battery and the fan.

9. The electric lawnmower control system according to claim 1, wherein the printed circuit board is located between the battery and the electric motor.

10. The electric lawnmower control system according to claim 1, wherein the housing contains the battery, the electric motor, the fan, and the printed circuit board.

11. The electric lawnmower control system according to claim 1, wherein the battery, the electric motor, the fan and the printed circuit board are located between the first side and the second side.

12. The electric lawnmower control system according to claim 1, wherein the first side and the second side extend along all of the battery, the electric motor, the fan and the printed circuit board.

13. The electric lawnmower control system according to claim 1, wherein the electric motor and the fan are located inside the housing the housing includes a front end and a rear end, the battery is mounted in the housing at a location that is adjacent to the front end and between the front end and the fan, the first side extends from the front end to the rear end, and the second side extends from the front end to the rear end.

14. An electric lawnmower comprising: the control system according to claim 1; a cutter housing; a shaft driven by the electric motor a plurality of wheels support the cutter housing; a handle connected to the cutter housing and extending away from the cutter housing; and at least one control input mounted on the handle and in electrical communication with the motor controller, wherein the electric motor is mounted on and supported by the cutter housing, the housing is mounted on the cutter housing, the blade is located in the cutter housing and connected to the shaft, and the fan is connected to the shaft.

15. The electric lawnmower according to claim 14, wherein the plurality of wheels includes a pair

of front wheels and a pair of rear wheels that are spaced away from the front wheels in a first direction, the ventilation passage includes an inlet on the first side of the housing and an outlet on the second side of the housing, and the outlet is spaced away from the inlet in a direction that intersects the first direction.

16. A control system for an electric lawnmower, comprising: a battery assembly including a battery case and at least one battery mounted in the battery case; an electric motor configured to rotatably drive a blade; a motor connector electrically connected to the electric motor; a fan rotatably driven by the electric motor; a printed circuit board mounted on the battery and including a heat sink that is located adjacent to the fan; a motor controller mounted on the circuit board, in electrical communication with the battery, thermally coupled to the heat sink, and configured to selectively drive the electric motor with electric power supplied by the battery; a first connector mounted on the printed circuit board and in electrical communication with the motor controller, physically and electrically connected to the motor connector, and in electrical communication with the motor controller; and a housing including a first side, a second side, and a ventilation passage extending from the first side to the second side, the ventilation passage is in fluid communication with the fan, and at least a portion of the heat sink is exposed to air flowing through the ventilation passage.

17. The control system according to claim 16, further comprising: a second connector mounted on the printed circuit board and in electrical communication with the motor controller; and a motor wire harness including at least one motor connector and a plurality of wires electrically connected to the motor and the motor connector, the motor connector is physically and electrically connected to the first connector, wherein the battery assembly includes a battery connector mounted on the battery case, and the battery connector is physically and electrically connected to the second connector.

18. A control system for an electric lawnmower, comprising: a battery assembly including a battery case and at least one battery mounted in the battery case; an electric motor configured to rotatably drive a blade; a fan housing; a fan located in the fan housing and rotatably driven by the electric motor; a motor controller in electrical communication with the battery and the electric motor and configured to selectively drive the electric motor with electric power supplied by the battery; and a heat sink thermally coupled to the motor controller and located adjacent to the fan; and a housing including a first side, a second side, and a ventilation passage extending from the first side to the second side, the ventilation passage is in fluid communication with the fan, the fan housing, and at least a portion of the heat sink, wherein the heat sink extends into the fan housing.

19. The control system according to claim 18, further comprising: a printed circuit board; a first connector mounted on the circuit board and in electrical communication with the motor controller; a second connector mounted on the circuit board and in electrical communication with the motor controller; a battery connector on the battery case, in electrical communication with the battery, and physically and electrically connected to the first connector; and a motor connector in electrical communication with the electric motor, and physically and electrically connected to the second connector.
